instituto superior de engenharia de lisboa

Master Thesis Title, Project Work or Internship Report

JOHN VERY LONGNAME DOE

(Graduated)

Dissertation plan submitted in partial fulfillment of the requirements for the degree of

MSc in Computer Science and Engineering

SUPERVISORS

Prof. Doctor Elisabeth Adviser Name Doe
Prof. Doctor John Adviser Name Doe Male

July of 2026

Master Thesis Title, Project Work or Internship Report

Abstract

The dissertation must contain two versions of the abstract, one in the same language as the main text, another in a different language. The package assumes that the two languages under consideration are always Portuguese and English.

The package will sort the abstracts in the appropriate order. This means that the first abstract will be in the same language as the main text, followed by the abstract in the other language, and then followed by the main text. For example, if the dissertation is written in Portuguese, first will come the summary in Portuguese and then in English, followed by the main text in Portuguese. If the dissertation is written in English, first will come the summary in English and then in Portuguese, followed by the main text in English.

The abstract should not exceed one page and should answer the following questions:

The format of your abstract will depend on the discipline in which you are working. However, all abstracts generally cover the following five sections:

- The purpose of the research (what's it about and why's that important)
- The methodology (how you carried out the research)
- The key research findings (what answers you found)
- The implications of these findings (what these answers mean)

Maximum of words: 300

Keywords: Keywords (in English) …– minimum of 3 (three)

Master Thesis Title, Project Work or Internship Report

Resumo

Independentemente da língua em que está escrita a dissertação, é necessário um resumo na língua do texto principal e um resumo noutra língua. Assume-se que as duas línguas em questão serão sempre o Português e o Inglês.

O *template* colocará automaticamente em primeiro lugar o resumo na língua do texto principal e depois o resumo na outra língua. Por exemplo, se a dissertação está escrita em Português, primeiro aparecerá o resumo em Português, depois em Inglês, seguido do texto principal em Português. Se a dissertação está escrita em Inglês, primeiro aparecerá o resumo em Inglês, depois em Português, seguido do texto principal em Inglês.

O resumo não deve exceder uma página e deve responder às seguintes questões:

- Qual é o problema?
- Porque é que ele é interessante?
- Qual é a solução?
- O que resulta (implicações) da solução?

Máximo de palavras: 300

Palavras-chave: Palavras-chave (em Português) …– mínimo de 3 (três)

Contents

List of Figures

List of Tables

Listings

List of Algorithms

Acronyms

α	First letter of the greek alphabet 11
β	Second letter of the greek alphabet 11
GCD	Greatest Common Divisor 11
LCM	Least Common Multiple 11

Glossary

matrix	a rectangular table of elements 11
universal set	the set of all things 10

1 INTRODUCTION

This package is distributed under GPLv3 License. If you have questions or doubts concerning the guarantees, rights and duties of those who use packages under GPLv3 License, please read http://www.gnu.org/licenses/gpl.html.

A margin-par note!

A a note in a line by itself.

Please note that

this package and template are not official for ISEL/IPL.

2 ISELTHESIS USERS MANUAL

2.1 Introduction

This chapter describes how to use the LaTeX style `thesis`. This style file is a major rewrite from the most of Universities, which was in turn adapted from a style file from the FCT-UNL [4]. We aimed at providing an improved visual layout and, simultaneously, a *very easy to use* template (aka, a LaTeX template for dummies).

The first main rule you must know is that **you must** specify the encoding of your text files. A simple *rule of thumb* is: if you are using Windows add 'latin1' to the list of package options; if you are using other systems, such as Linux or Mac OSx, add 'utf8' to the list of package options.

2.2 Folder Structure

The template file for writing dissertations in LaTeX is organised into a main directory, a set of files and sub-directories:

iselthesis This is the main directory and includes:

Chapters Directory where to put user files (text and figures). For instances, including Chapters, Dedicatory, Acknowledgments, Gantt Diagram, Appendices, Annexes[1], Abstracts, Glossaries, Lists of Symbols, etc. Replace them with your own. The "Statement" file is important for submitting the final version of the master's thesis. It contains the text in both languages (English and Portuguese). The author must choose one of these depending on whether they have written their dissertation in English or Portuguese respectively. This folder includes:

1. **scripts** Directory where you can place the code you need to insert into the text, or pseudo-code;
2. **img** Directory with all images of your thesis;

[1]**Note**: Change the file name if you don't want to print. Like, instead of "ganttdiagram.tex" choose to "ganttdiagram_.tex", for instance

Config Identify the directory where you will customise your content, including setting up chapters, appendices and appendices, and entering any personal information:

1. **_cover.tex** Configure the contents of the cover: e.g. title of the thesis, author's name, names of committee members (including supervisor), date and figure;
2. **_institution.tex** Fixed institutional data (university and school names). Only change this file if you are **not** from ISEL;
3. **_files.tex** Select the files for chapters, appendices, annexes, abstracts, glossaries, etc;
4. **_packages.tex** User customisation. Load additional packages and define your own commands to be used throughout the documentation.

Logo Directory with Faculty logos;

Scripts Directory with useful bash scripts, e.g., for cleaning all temporary files;

alpha-pt.bst A file with bibliography names in portuguese, e.g., 'Relatório Técnico' e 'Tese de Mestrado' instead of 'Technical Report' and 'Master Thesis'. This file is used automatically if Portuguese is selected as the main language (see below);

template.tex The main file. You should run LaTeX in this one. Please refrain from changing the file content outside of the well defined area;

bibliography.bib The bib file. An easy way to find to import citation into `bibtex` is select option `Show links to import citation into BibTex` in `Scholar google settings`.

iselthesis.cls The LaTeX class file for the thesis style. This file should not be changed, unless you're ready to play with fire! :)

Again, we would like to recall that all the user LaTeX files should be stored in the `iselthesis` directory, and all the images in `iselthesis/Chapters/img` directory.

Yet another note!

2.3 Package Options

The thesis style includes the following options (see Table 2.1), that must be included in the options list in the `\documentclass[options]` line at the top of the `template.tex` file.

The list below aggregates related options in a single item. For each list, the default value is prefixed with a asterisk (*).

Table 2.1. Description of options used in this document

Options	Definition and values
docdegree	**Document degree**: Master – msc(*), Preparation of Master – mscprep and Bachelor – bsc degree.
doctype	**Type of document**: dissertation(*) or project.
lang	**The main language for the document**: English – en(*) and Portuguese – pt.
coverlang	**The language to be used when typesetting the cover page**: English – en(*) and Portuguese – pt.
fontstyle	**The font set to be used in the document**: baskervaldx bookman charter ebgaramond fbb fourier garamond heuristica kpfonts libertine mathpazo1 mathpazo2 newcent newpx newtx verdana(*)
chapstyle	**The chapter style to be used in the document**: bianchi bluebox brotherton dash default elegant ell ger hansen isel(*) jenor lyhne madsen pedersen veelo vz14 vz34 vz43
otherlistsat	**Place where to put the other lists besides the table of contents**: front(*) and back.
linkscolor	**The color for all the hyperlinks in the PDF file**: darkblue, black (Set to 'black' for PRINTING).
printcommittee	**List of the committee members**: set to 'false' from submitted versions who should not have the list of committee members.
biblatex	**Customize** `biblatex`, the bibliography management system used in this class: list of options.
memoir	**Customize the base class** `memoir`: list of options.
media	**The target of the PDF**: screen(*) or paper.

2.3.1 Language Related Options

You must choose the main language for the document. The available options are:

1. **pt** — The document is written in Portuguese (with a small abstract in English).
2. **en** — The document is written in English (with a small abstract in Portuguese).

The language option affects:

- **The order of the summaries.** At first the abstract in the main language and then in the foreign language. This means that if your main language for the document in english, you will see first the abstract (in english) and then the 'resumo' (in portuguese). If you switch the main language for the document, it will also automatically switch the order of the summaries.

- **The names for document sectioning.** E.g., 'Chapter' vs. 'Capítulo', 'Table of Contents' vs. 'Índice', 'Figure' vs. 'Figura', etc.

- **The type of documents in the bibliography.** E.g., 'Technical Report' vs. 'Relatório Técnico', 'MSc Thesis' vs. 'Tese de Mestrado', etc.

No mater which language you chose, you will always have the appropriate hyphenation rules according to the language at that point. You always get portuguese hyphenation rules in the 'Resumo', english hyphenation rules in the 'Abstract', and then the main language hyphenation rules for the rest of the document. If you need to force hyphenation write inside of `\hyphenation{}` the hyphenated word, e.g. `\hyphenation{op-ti-cal net-works}`.

2.3.2 Class of Text

You must choose the class of text for the document. The available options are:

1. **bsc** — BSc graduation report.

2. **mscprep** — Preparation of MSc dissertation. This is a preliminary report graduate students at ISEL/IPL must prepare to conclude the first semester of the two-semesters MSc work. The files specified by a) `\dedicatoryfile` and b) `\acknowledgmentsfile` are ignored, even if present, for this class of document.

3. **msc** — MSc dissertation.

2.3.3 Printing

You must choose how your document will be printed. The available options are:

1. **media**(*) — Single side page printing, and

2. **paper** — Double sided page printing.

The article 50th, of Decree-Law No. 115/2013, requires the deposit of a digital copy of doctoral thesis and master's dissertations in a repository that is part of the RCAAP repository[2], https://www.rcaap.pt. This deposit aims to preserve scientific work, as well as providing Open Access to scientific production is not restricted object or embargo.

[2]Repositórios Científicos de Acesso Aberto de Portugal

For the reason explained above, we include the option to format your thesis in a way that presents well on screen and/or on paper. But always remember that your work will be stored in the RCAAP portal in electronic format.

2.3.4 Text Encoding

You must choose the font size for your document. The available options are:

1. **latin1** — Use Latin-1 (ISO 8859-1) encoding. Most probably you should use this option if you use Windows;
2. **utf8** — Use UTF8 encoding. Most probably you should use this option if you are not using Windows.

2.3.5 Examples

Let's have a look at a couple of examples:

- BSc graduation report, in portuguese, considering printing on screen)
  `documentclass[docdegree=bsc, lang=pt, media=screen]{iselthesis}`
- Preparation of MSc thesis, in portuguese, with link color equal to black (I wonder why one would do this!). Note that, `pt` is declared by default, so it can be omitted:
  `documentclass[docdegree=mscprep, lang=pt, linkcolor=black]{iselthesis}`
- MSc's project, in english and to be printed single sided on screen. Note that, `twoside` and `12pt` are declared by default, so it can be omitted:
  `documentclass[docdegree=msc, doctype=project, lang=en,`
  `media=screen]{iselthesis}`

The present document is defined according to the following settings:

```
documentclass[
  docdegree=msc,          % msc(*), mscprep, bsc
  doctype=dissertation, % dissertation(*), project
  lang=pt,                % en, pt(*)
  coverlang=pt,           % defaults to main language
  fontstyle=verdana,      % baskervaldx ... verdana(*)
  chapstyle=isel,         % bianchi ... isel(*) jenor vz43
  otherlistsat=front,     % front(*), back
  linkscolor=darkblue,    % darkblue(*), black
  printcommittee=true,
  biblatex={   % Options for biblatex (see biblatex documentation)
        sorting=nyt,          % none, nyt(*), ynt
        backref=true,         % true(*), false
    maxbibnames=99,                       % Never use 'et al'
```

```
    hyperref=true       % Hyperlinks in citations: true(*) false
        },
  memoir={                    % See the 'memoir' documentation
    a4paper,            % the paper size/format
    11pt,               % 10pt, 11pt(*), 12pt
    final,              % draft, final
    },
  media=screen,         % screen(*), paper
]{iselthesis}
```

2.4 How to Write Using LaTeX

Please have a look at Chapter 3, where you may find many examples of LaTeX constructs, such as sectioning, inserting figures and tables, writing equations, theorems and algorithms, exhibit code listings, etc.

3 A SHORT LaTeX TUTORIAL WITH EXAMPLES

This Chapter aims at exemplifying how to do common stuff with LaTeX. We also show some stuff which is not that common! ;)

Please, use these examples as a starting point, but you should always consider using the *Big Oracle* (aka, Google, your best friend) to search for additional information or alternative ways for achieving similar results.

3.1 Document Structure

In engineering and science, a thesis or dissertation is the culmination of a master's or Ph.D. degree. A thesis or dissertation presents the research that the student performed for that degree. From the student's perspective, the primary purpose of a thesis or dissertation is to persuade the student's committee that he or she has performed and communicated research worthy of the degree. In other words, the main purpose of the thesis or dissertation is to help the student secure the degree.

From the perspective of the engineering and scientific community, the primary purpose is to document the student's research. Although much research from theses and dissertations is also communicated in journal articles, theses and dissertations stand as detailed documents that allow others to see what the work was and how it was performed. For that reason, theses and dissertations are often read by other graduate students, especially those working in the research group of the authoring student [1, 2, 6, 7].

With a thesis or dissertation, the format also encompasses the names of the sections that are expected:

1. Thesis Cover
2. Acknowledgments (if exist)
3. Abstract (Portuguese and English)
4. Index
5. List of Figures
6. List of Tables

7. Listings
8. List of Algorithms
9. Nomenclature/List of Abbreviations (if exists)
10. Glossary (if exists)
11. Introduction
12. State-of-the-Art or Related work
13. Proposed method
14. Experiment result
15. Conclusion and Future work
16. References,
17. Gantt diagram (if exists), and
18. Appendix (if exists)

3.1.1 State-of-the-Art

State-of-the-Art (SoTA) is a step to demonstrate the novelty of your research results. The importance of being the first to demonstrate research results is a cornerstone of the research business[1].

Besides demonstrating the novelty of your research results, a SoTA has other important properties:

1. It teaches you a lot about your research problem. By reading literature related to your research problem you will learn from other researchers and it will be easier for you to understand and analyse your problem;
2. It proves that your research problem has relevance;
3. It shows different approaches to a solution;
4. It shows what you can reuse from what others have done.

3.1.2 Related work

In the *Related Works* section, you should discuss briefly about published matter that technically relates to your proposed work[2].

A short summary of what you can include (but not limited to) in the Related Works section:

[1] "Why and how to write the state-of-the-art", by Babak A. Farshchian, May 22, 2007

[2] https://academia.stackexchange.com/questions/68164/how-to-write-a-related-work-section-in-computer-science

1. Work that proposes a different method to solve the same problem;
2. Work that uses the same proposed method to solve a different problem;
3. A method that is similar to your method that solves a relatively similar problem;
4. A discussion of a set of related problems that covers your problem domain.

3.2 Glossary and Nomenclature/List of Symbols

Many technical documents use terms or acronyms unknown to the general population. It is common practice to add a glossary to make such documents more accessible. A *glossary* is a nice thing to have in a report and usually very helpful. As you probably can imaging, it is very easy to create in LaTeX.

As with all packages, you need to load glossaries with `\usepackage`, but there are certain packages that must be loaded before glossaries, if they are required: `hyperref`, `babel`, `polyglossia`, `inputenc` and `fontenc`.

```
\usepackage{glossaries}
\makenoidxglossaries
```

Once you have loaded `glossaries`, you need to define your terms in the preamble (or, separated file) and then you can use them throughout the document.

Next you need to define the terms you want to appear in the glossary. Again, this must be done in the preamble. This is done using the command

`\newglossaryentry{<label>}{<key-val list>}`

The first argument `<label>` is a unique label to allow you to refer to this entry in your document text. The entry will only appear in the glossary if you have referred to it in the document using one of the commands listed later. The second argument is a comma separated `<key>=<value>` list.

Inside the text you just need to use the command `\gls{name}` or `\glspl{name}` (plural name) to call it. For example, the following defines the term 'set' and assigns a brief description. The term is given the label set. This is the minimum amount of information you must give:

```
\newglossaryentry{set} % the label
{ name=set,                    % the term
  description={a collection of objects}  % a brief description
}
```

Other example, now the glossary associated with a symbol, universal set:

```
\newglossaryentry{U}   % the label
{ name={universal set}, % the term
```

```
  description={the set of all things}   % a brief description
  symbol={\ensuremath{\mathcal{U}}}   % the associated symbol
}
```

Here's a simple example:

```
\usepackage{glossaries}
\newglossaryentry{ex}{name={sample},description={an example}}
\newacronym{svm}{SVM}{support vector machine}
\newacronym{beta}{$\beta$}{Second letter of the greek alphabet}
\newacronym{alpha}{$\alpha$}{First letter of the greek alphabet}

\begin{document}
Here's my \gls{ex} term. First use: \gls{svm}.
Second use: \gls{svm}.

\textit{I want the \gls{beta} to be listed after the \gls{alpha}}.
\end{document}
```

This produces: *Here's my sample term. First use: support vector machine (SVM). Second use: SVM.*
I want the Second letter of the greek alphabet (β) to be listed after the First letter of the greek alphabet (α).

Do not use `\gls` in chapter or section headings as it can have some unpleasant side-effects. Instead use `\glsentrytext` for regular entries and one of `\glsentryshort`, `\glsentrylong` or `\glsentryfull` for acronyms. Alternatively use glossaries-extra which provides special commands for use in section headings, such as `\glsfmtshort{<label>}`.

The plural of the word "matrix" is "matrices" not "matrixs", so the term needs the plural form set explicitly:

```
\newglossaryentry{matrix}% the label
{ name=matrix, % the term
  description={a rectangular table of elements},
 plural=matrices % the plural
}
```

Given a set of numbers, there are elementary methods to compute its Greatest Common Divisor, which is abbreviated GCD. This process is similar to that used for the Least Common Multiple (LCM).

3.3 Importing Images

You can import external graphics using package `graphicx`. You will find several examples here: https://www.overleaf.com/learn/latex/Inserting_Images. The most important command is `\includegraphics`. LaTeX itself treats the image like normal text, i.e. as a box of certain height and width. The package documentation list the options width and height, as well as others.

This syntax is fairly self-explanatory, however, one thing to note is the use of the `\centering` command. There are other ways to center a figure, but this is the correct way to do it inside a float environment. You will almost always be using this command when using floats.

```
\begin{figure}[H]
\centering
\includegraphics[scale=1.5]{atom.png}
\end{figure}
\end{document}
```

3.4 Floats Figures and Tables, and Captions

The tabular environment can be used to typeset tables with optional horizontal and vertical lines. LaTeX determines the width of the columns automatically. The first line of the environment has the form: `\begin{tabular}[pos]{table spec}`, where (1) `table spec` tells LaTeX the alignment to be used in each column and the vertical lines to insert; and, (2) `pos` can be used to specify the vertical position of the table relative to the baseline of the surrounding text. The number of columns does not need to be specified as it is inferred by looking at the number of arguments provided. It is also possible to add vertical lines between the columns here. You can draw beautiful tables with the `booktabs` package https://ctan.org/pkg/booktabs?lang=en.

Some notes are important to followed, such as present in Table 3.1:

i) Not defined vertical lines;

ii) The legend must be on top;

iii) Use `\toprule`, `\midrule` and `\bottomrule` to draw horizontal lines.

Table 3.1. Table's rules.

Item		
Animal	Description	Price ($)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

There are two ways to incorporate images into your LaTeX document, and both use the graphicx package by means of putting the command `\usepackage{graphicx}` near the top of the LaTeX file, just after the documentclass command.

The two methods are

1. include only PostScript images (esp. 'Encapsulated PostScript') if your goal is a PostScript document using dvips;

2. include only PDF, PNG, JPEG and GIF images if your goal is a PDF document using pdflatex, `TeXShop`, or other PDF-oriented compiler.

Some PNG images within my LaTeX document. The quality of the image files is sufficient and the result using LaTeX and viewing the resulting DVI file is quite looks good.

To get the best quality of the images in PDF files I'd recommend using vector-based graphics for images. The best format to save images in is .pdf, see Figure 3.1a. With programs like `Inkscape`, you can draw as you would in MS Paint (and do much more), and because the images are vector-based instead of pixel-based, their quality should be preserved when converting to PDF in any way.

In all cases, each image must be in an individual 1-image file; no animation files or multipage documents.

There are two different ways to place two figures/tables side-by-side. More complicated figures with multiple images. You can do this using subcaption environments[3] inside a figure environment. Subcaption will alphabetically number your figures and you have access to the complete reference as usual through `\ref{fig:figurelabel}`, Figure 3.1, or Figure 3.1b using `\ref{fig:subfigurelabel}`.

Using the package listings you can add non-formatted text as you would do with `\begin{verbatim}` but its main aim is to include the source code of any programming language within your document. If you wish to include pseudo-code or algorithms see LaTeX/Algorithms_ and_Pseudocode, as Algorithm 3.1.

```
# comentário
square <- function(x) {
    x^2
    % |$x^{2}$|
}

# nerv
x <- c(1:100)
y <- square(x)
```

Listing 3.1. R-Code (Test).

[3] The subcaption package: http://mirrors.ibiblio.org/CTAN/macros/latex/contrib/caption/subcaption.pdf

(a) *Vectorial*

(b) *Raster*

Figure 3.1. *Subfigure example with vectorial and no-vectorial images*

Algorithm 3.1 An algorithm with caption

Require: $n \geq 0$
Ensure: $y = x^n$
  $y \leftarrow 1$
  $X \leftarrow x$
  $N \leftarrow n$
  while $N \neq 0$ **do**
    if N is even **then**
      $X \leftarrow X \times X$
      $N \leftarrow \frac{N}{2}$ ▷ This is a comment
    else if N is odd **then**
      $y \leftarrow y \times X$
      $N \leftarrow N - 1$
    end if
  end while

3.5 Generating PDFs from LaTeX

3.5.1 Generating PDFs with pdflatex

You may create PDF files either by using `latex` to generate a DVI file, and then use one of the many DVI-2-PDF converters, such as `dvipdfm`.

Alternatively, you may use `pdflatex`, which will immediately generate a PDF with no intermediate DVI or PS files. In some systems, such as Apple, PDF is already the default format for LaTeX . I strongly recommend you to use this approach, unless you have a very good argument to go for `latex` + `dvipdfm`.

A typical pass for a document with figures, cross-references and a bibliography would be:

```
$ pdflatex template
```

```
$ bibtex template
$ pdflatex template (twice)
```

You will notice that there is a new PDF file in the working directory called `template.pdf`. Simple :)

Please note that, to be sure all table of contents, cross-references and bibliography citations are up-to-date, you must run `latex` once, then `bibtex`, and then `latex` twice.

3.5.2 Dealing with Images

You may process the same source files with both `latex` or `pdflatex`. But, if your text include images, you must be careful. `latex` and `pdflatex` accept images in different (exclusive) formats. For `latex` you may use EPS ou PS figures. For `pdflatex` you may use JPG, PNG or PDF figures. I strongly recommend you to use PDF figures in vectorial format (do not use bitmap images unless you have no other choice).

3.5.3 Creating Source Files Compatible with both latex and pdflatex

Do not include the extension of the figure file file in the `includegraphics` command, for instance use: `\includegraphics{filename}`, and not: `\includegraphics{filename.png}`. In the first form, `latex` or `pdflatex` will add an appropriate file extension.

This means that, if you plan to use only `pdflatex`, you need only to keep (preferably) a PDF version of all the images. If you plan to use also `latex`, then you also need an EPS version of each image.

To be included in the sections above

If you are writing only one or two documents and aren't planning on writing more on the same subject for a long time, maybe you don't want to waste time creating a database of references you are never going to use. In this case you should consider using the basic and simple bibliography support that is embedded within LaTeX .

LaTeX provides an environment called `thebibliography` that you have to use where you want the bibliography; that usually means at the very end of your document, just before the `\end{document}` command. Here is a practical example:

```
\begin{thebibliography}{9}
\bibitem{lamport94}
  Leslie Lamport,
  \emph{\LaTeX: A Document Preparation System}.
  Addison Wesley, Massachusetts,
  2nd Edition,
  1994.
\end{thebibliography}
```

In this document, the bibliography is in a separate document: `bibliography.bib` where information is entered from `https://scholar.google.pt/`, as show Figure 3.2.

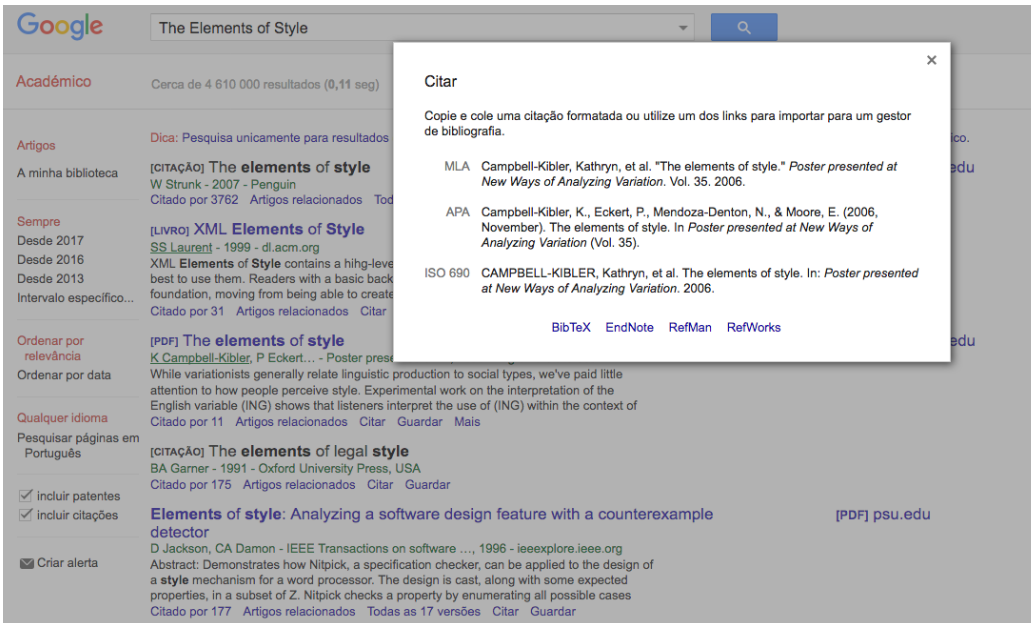

Figure 3.2. *Screenshot from Scholar Google*

To actually, cite a given document is *very* easy. Go to the point where you want the citation to appear, and use the following: `\cite{citekey}`, where the `citekey` is that of the `bibitem` you wish to cite, e.g `~\cite{lamport94}`. When LaTeX processes the document, the citation will be cross-referenced with the bibitems and replaced with the appropriate number citation. The advantage here, once again, is that LaTeX looks after the numbering for you.

When a sequence of multiple citations are needed, you should use a single `\cite{}` command. The citations are then separated by commas. Note that you must not use spaces between the citations. Here's an result example [3, 6, 7].

Footnotes are a very useful way of providing extra information to the reader. Usually, it is non-essential information which can be placed at the bottom of the page. This keeps the main body of text concise.

The footnote facility is easy to use: `\footnote{Simple footnote}`[4].

3.6 Equations

Typesetting mathematics is one of LaTeX's greatest strengths. It is also a large topic due to the existence of so much mathematical notation. It is recommend to read the following document available in Short Math Guide for LaTeX - AMS - American Mathematical Society.

[4] Simple footnote

3.7 Page orientation

The default page layout is “portrait”, but sometimes it is still useful/necessary to have the whole document or only single pages changed to “landscape”. The latter might be due to a large table or figure. If you want to make appear the left side up, better readable on screen, the pdflscape-package will do it: `\usepackage{pdflscape}`

and again:

```
\begin{landscape}
...
\end{landscape}
```

or, `\includepdf[landscape=true,pages={1}]{example.pdf}`

to put the page in “landscape”, while the rest will remain in “portrait” orientation. Nevertheless, the header/footer will also be changed in orientation.

Written by Matilde Pós-de-Mina Pato with collaboration of Nuno Datia, 2012 October (1st version)

4 LOREM IPSUM

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

Bibliography

[1] H. Glasman-Deal. *Science research writing for non-native speakers of English*. World Scientific, 2010 (cit. on p. 8).

[2] B. Gustavii. *How to write and illustrate a scientific paper*. Cambridge University Press, 2016 (cit. on p. 8).

[3] D. E. Knuth. *The TeXbook*. Addison-Wesley Publisher, 1984 (cit. on p. 16).

[4] J. M. Lourenço. *The NOVAthesis LaTeX Template User's Manual*. NOVA University Lisbon. 2021. URL: https://github.com/joaomlourenco/novathesis/raw/main/template.pdf (cit. on p. 2).

[6] W. Strunk and R. D. A'Morelli. *The Elements of Style: Classic Edition*. 1st edition. Spectrum Ink Publisher, 2018. ISBN: 978-1-64399-000-2 (cit. on pp. 8, 16).

[7] G. Webster. *The Chicago Manual of Style*. 17th edition. University of Chicago Press, 2017, pp. 400–401. ISBN: 978-0-226-28705-8 (cit. on pp. 8, 16).

A APPLIED SURVIVAL ANALYSIS BY HOSMER AND LEMESHOW

Stata Textbook Examples Applied Survival Analysis by Hosmer and Lemeshow
The data files used for the examples in this text can be downloaded in a zip file from the Wiley FTP website or the Stata Web site.

```r
# The R package(s) needed for this chapter is the survival package.
# We currently use R 2.0.1 patched version. You may want to make sure
# that packages on your local machine are up to date. You can perform
# updating in R using update.packages() function.

# url: http://www.ats.ucla.edu/stat/r/examples/
# data set is hmohiv.csv.
hmohiv<-read.table("http://www.ats.ucla.edu/stat/r/examples/asa/hmohiv.csv", sep=","
    , header = TRUE)
attach(hmohiv)
hmohiv

# using the hmohiv data set. To control the type of symbol, a variable called
    psymbol is created.
# It takes value 1 and 2, so the symbol type will be 1 and 2.
psymbol<-censor+1
table(psymbol)

plot(age, time, pch=(psymbol))
legend(40, 60, c("Censor=1", "Censor=0"), pch=(psymbol))

age1<-1000/age
plot(age1, time, pch=(psymbol))
legend(40, 30, c("Censor=1", "Censor=0"), pch=(psymbol))

# Package "survival" is needed for this analysis and for most of the analyses in the
    book.
library(survival)
test <- survreg( Surv(time, censor) ~ age, dist="exponential")
summary(test)

pred <- predict(test, type="response")
ord<-order(age)
age_ord<-age[ord]
pred_ord<-pred[ord]
plot(age, time, pch=(psymbol))
lines(age_ord, pred_ord)
legend(40, 60, c("Censor=1", "Censor=0"), pch=(psymbol))
```

B APPENDIX 2 LOREM IPSUM

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

I ANNEX 2 LOREM IPSUM

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.